Does a moving viewpoint change the exoplanet sky?

An illustrated report from Finding Earth 2.0

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1. Motivation

I began with a map of known exoplanet hosts and a question: if the Solar System moves through the Milky Way, does the exoplanet sky accessible to us change? The original discussion note describes an observer moving from A to B and then C while target systems move too.

A changing sightline can matter even when a target stays far away. A planet close to the edge of a transit band might enter or leave it. Distance changes can also alter angular separation and brightness. These effects must be distinguished from improvements in telescopes.

This report brings together that original question, a reproducible simulation and an exploratory Gaia-based calculation. It asks for scientific criticism; it does not claim a new physical law or a forecast of habitable-planet discoveries.

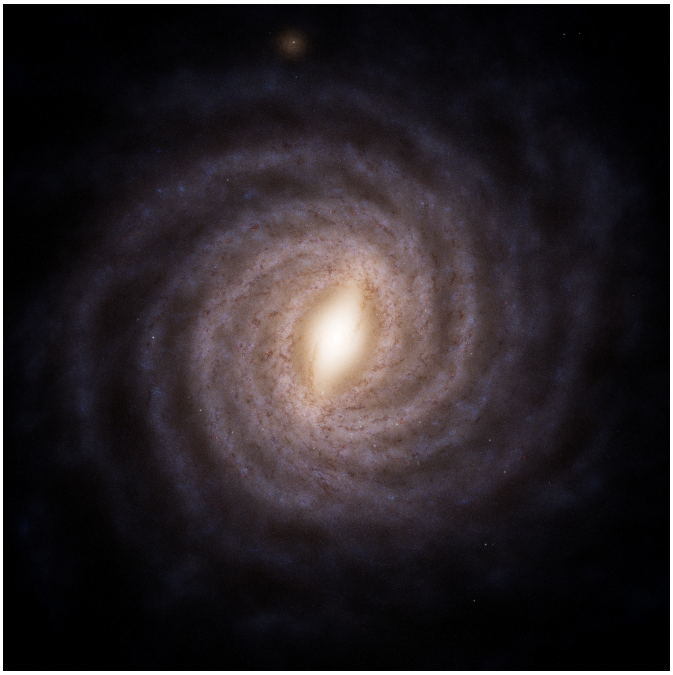


Figure 1. Gaia-informed artist reconstruction of the Milky Way, not an external photograph or a future star-count map. ESA/Gaia/DPAC, Stefan Payne-Wardenaar; CC BY-SA 3.0 IGO [1].

A moving observer and a moving target

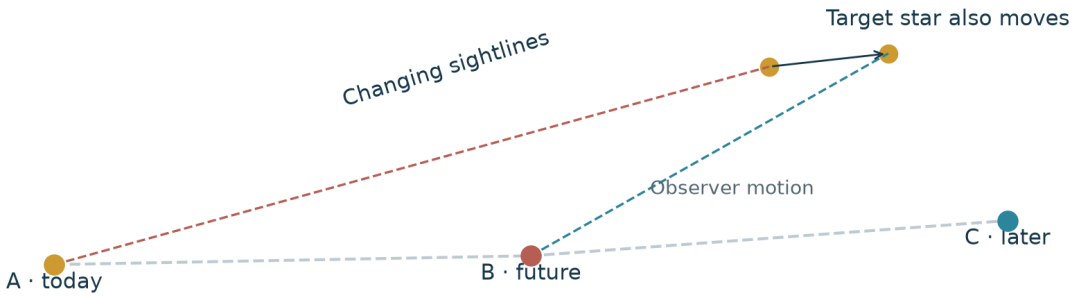


Figure 2. Conceptual A-to-B-to-C observer epochs and two target positions. Dashed sightlines change because both ends move. The drawing is schematic; A/B/C here are not the optional synthetic comparison stars in the website.

Central question: for which systems, observing methods and timescales does relative motion change a measurable opportunity to detect or study a planet?

2. What changes geometrically?

Let the observer position be the Solar position plus Earth's heliocentric offset. For a target star i, its instantaneous direction is determined by the difference of the two positions, not the observer's Galactic displacement alone.

$$\mathbf{r}_{\mathrm{obs}}(t) = \mathbf{r}_{\odot}(t) + \mathbf{r}_{\oplus/\odot}(t)$$

$$\hat{\mathbf{n}}_i(t) = \frac{\mathbf{r}_i(t) - \mathbf{r}_{\mathrm{obs}}(t)}{|\mathbf{r}_i(t) - \mathbf{r}_{\mathrm{obs}}(t)|}, \qquad \Delta\theta \simeq \frac{|\mathbf{v}_{\mathrm{rel},\perp}|\Delta t}{D}$$

Here D is the initial target distance and $v_{rel,perp}$ is the relative transverse velocity. The angular approximation assumes a small change and nearly constant relative velocity. Annual parallax and long-term proper motion are different contributions to a changing view.

A near-boundary transit example



Figure 3. Sky-projected planet tracks outside and inside a stellar disk. The planet is enlarged to clarify the geometry; these are not calculated future positions of a known system. A transit requires alignment as well as the correct orbital phase.

$$b(t) = \frac{a\cos i(t)}{R_\star}, \quad |b| < 1 + \frac{R_p}{R_\star}, \quad P_{\mathrm{transit}} \simeq \frac{R_\star + R_p}{a}$$

For a circular orbit, a is orbital radius, i is inclination to the line of sight, and R denotes radius. The probability averages isotropic orientations. For an Earth-Sun analogue it is about 0.47%; that does not mean 0.47% of all stars will yield a detected habitable planet.

Time interval	Sun at 233 km/s	Sightline shift: 30 km/s transverse at 100 ly
100 years	0.0777 light-years	20.6 arcseconds
10,000 years	7.77 light-years	0.573 degrees
100,000 years	77.7 light-years	5.73 degrees (linear estimate)

The last angular estimate is only a scale illustration: changing distance, acceleration and binary motion limit long extrapolations. Radial-velocity amplitude depends on sin i; direct imaging depends on angular separation, contrast and flux. Each method needs its own detection model.

Leaving a probe: a released spacecraft inherits its velocity and follows gravity. It does not stay at a fixed Galactic coordinate. The simulation's fixed 2026 marker is a hypothetical reference, not a passive parked probe.

3. What the catalogue experiment says

Changing Earth-transit-zone membership is established research [2]. The implemented experiment tests a narrower observer-side question: in a fixed nearby-source sample, how much does a century of relative motion change a single-visit direct-imaging accessibility score?

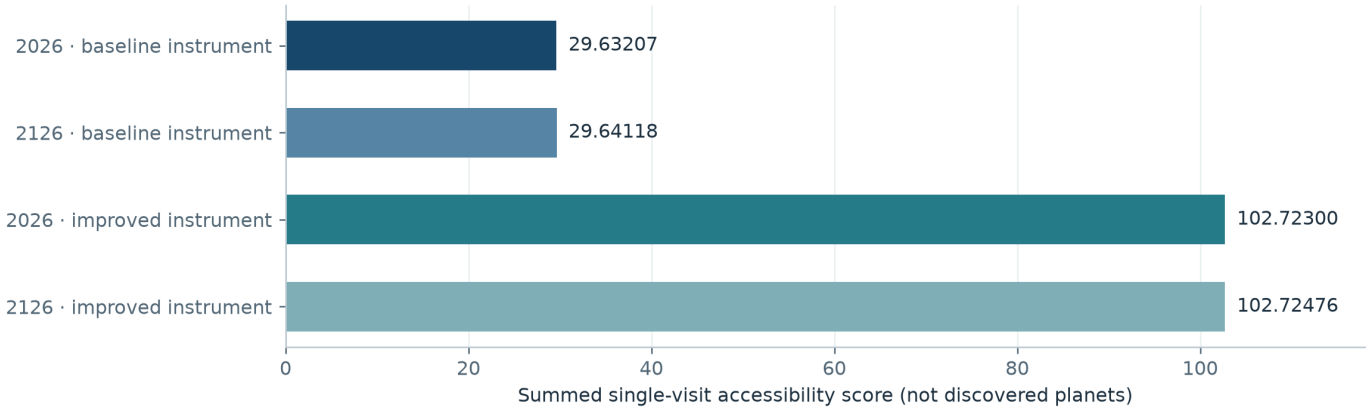


Figure 4. Four paired scenarios from the saved Gaia-based calculation. The two baseline bars differ by only 0.00911; the instrument change dominates visually. The horizontal axis is an accessibility sum, not detected-planet yield.

Data and assumptions

The selected Gaia DR3 sample [8] contains **1,387 sources** with parallax at least 50 mas, parallax signal-to-noise above 20, measured radial velocity, RUWE below 1.4 and at least ten visibility periods. It is not a complete census of nearby stars.

Each source receives a hypothetical Earth-radius planet on a circular 1 AU orbit with albedo 0.3. Positions are averaged over isotropic orientations and phases. Baseline thresholds are 60 mas inner working angle, contrast 10^{-10} and G = 30; the improved scenario uses 40 mas, 5×10^{-11} and G = 31.

Result and uncertainty

At fixed baseline thresholds, the score changes from **29.63207 to 29.64118**: about **+0.031%**. The assumed instrument improvement adds 73.09093 at the initial epoch. These settings are sensitivity scenarios, not approved mission specifications.

With 1,024 correlated astrometric draws per source, the motion difference has a formal 95% interval of +0.00868 to +0.01152. This interval excludes population uncertainty, instrument uncertainty, binary-system errors and cross-source systematics.

A useful counterexample

Forty-two sources gain and 45 lose accessibility at the stated thresholds; those are source-level score changes, not planet discoveries. Changing only the inner working angle to 70 mas produces a net motion contribution of about **-0.00145**. Waiting does not necessarily improve aggregate access.

What remains untested: averaging over isotropic planet positions removes sensitivity to absolute sky direction. This experiment therefore does not answer the original transit-boundary question for known orbital planes. A targeted, orbit-aware study is the next step.

4. Are we entering a different neighbourhood?

Many disk stars share Galactic rotation, but not identical velocities. Relative velocities determine encounters. Galactic radius R and height z describe different aspects of the environment; neither alone fixes an individual star's speed.

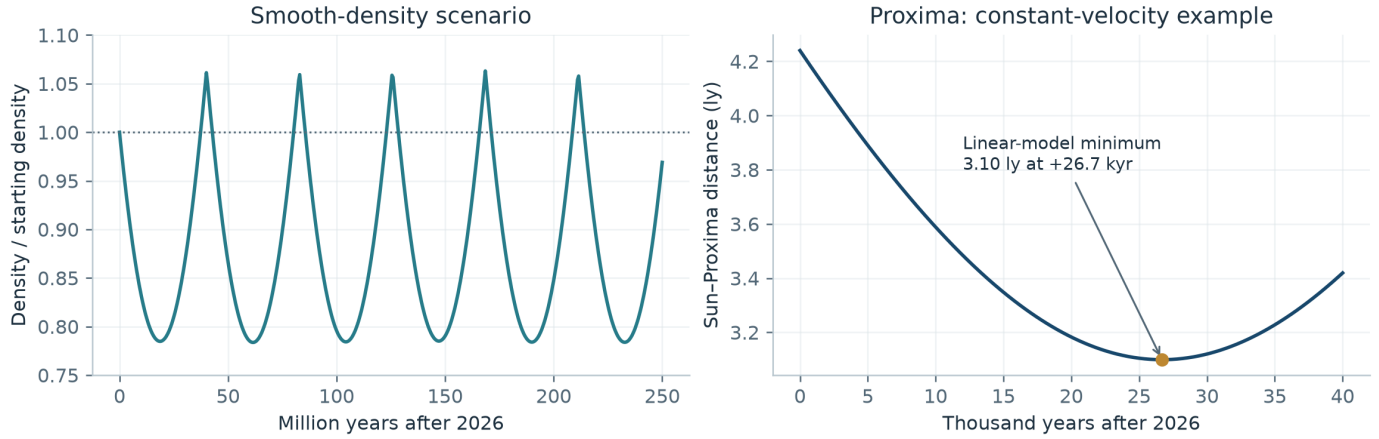


Figure 5. Left: a smooth thin/thick-disk density scenario along the integrated Solar orbit. Right: Proxima's relative trajectory under constant velocity using historical published inputs [3]. These independent models have different time ranges; the right panel is not a precise encounter forecast.

Density is not a discovery count

The scenario is normalized to a historical count of 373 stars within 10 pc [4]. Applying the adopted thin/thick-disk profile [5] gives roughly **0.784-1.063 times** the starting density over 250 Myr, or about 292-397 stars in an instantaneous 10 pc sphere.

These are not distinct arrivals or planets found. Spiral-arm enhancements, clusters and migration are omitted. The density profile is separate from the force model and from the artist reconstruction.

Proxima is approaching now

The adopted Proxima inputs yield about **4.24 ly** in 2026 and radial velocity **-22.19 km/s**: the distance is decreasing. The total relative speed is about 32.51 km/s, including transverse motion.

Straight-line extrapolation reaches about 3.10 ly after 26,700 years. The calculation omits the Alpha Centauri system's orbital dynamics and Galactic differential acceleration, so the minimum is illustrative.

Speed farther from the Galactic centre

Radius R	Model circular speed	Model orbital period
5 kpc	221.8 km/s	138.5 Myr
8.2 kpc	233.0 km/s	216.2 Myr
12 kpc	244.9 km/s	301.0 Myr
16 kpc	255.0 km/s	385.4 Myr

The existing potential rises outward in circular speed; it is **not a fit to the latest measured rotation curve**. Observations depend on tracers and assumptions [6]. Higher-z populations can have different random motions and mean rotation; there is no universal speed-height rule.

5. Reading the 3D simulation correctly

A trajectory's shape depends on the reference frame. The same planetary motion can appear as closed orbits in a co-moving frame and as winding paths when Solar travel is included. That does not imply a universal helix through spacetime.

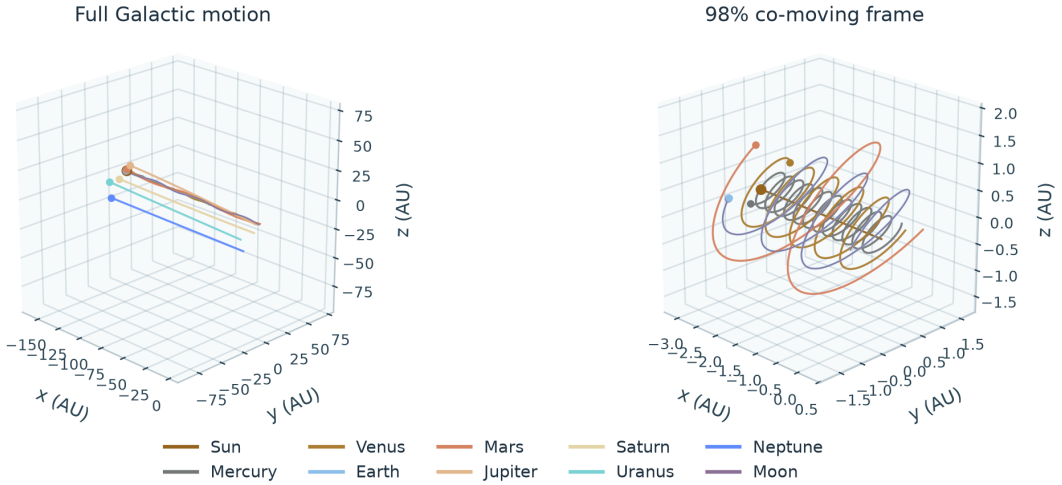


Figure 6. Numerical model rendering of three years of trajectories. Left: full Galactic translation. Right: a frame translating at 98% of the initial Solar velocity, resolving the inner-system winding. Each panel has equal AU axes but a different range; markers are enlarged. Not website screenshots.

The Sun's planetary reflex motion, its long-term vertical oscillation and the illustrative Galactic warp are different effects. The long clock uses an idealized nearly circular orbit in a static smooth potential. **Evolution of the Galactic plane is not modelled**; the animated warp does not drive the orbit.

Earth: gravity, motion and clocks

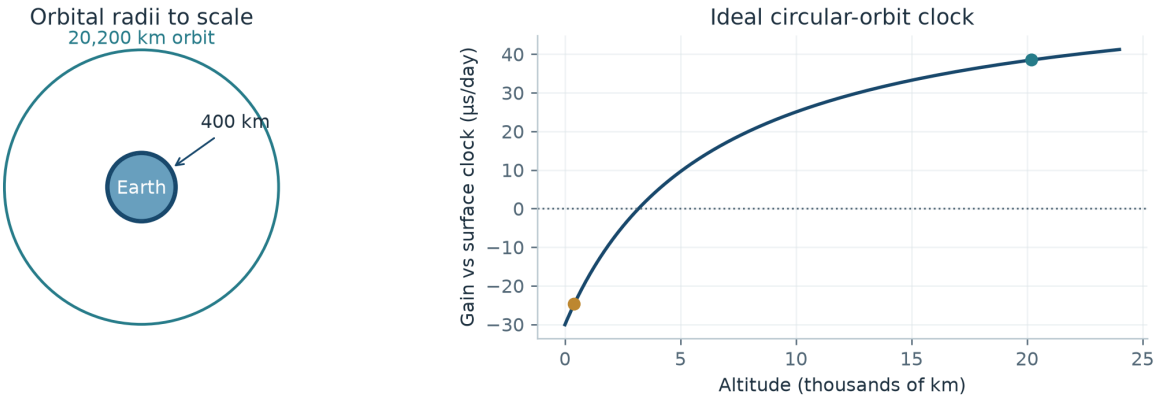


Figure 7. Left: ideal circular orbit radii relative to Earth's radius. Right: orbital-clock gain against a stationary surface clock for an isolated, spherical, nonrotating Earth. The 400 km case loses 24.7 μs/day; the 20,200 km case gains 38.5 μs/day.

$$\left(\frac{d\tau}{dt}\right)_{\text{held}} = \sqrt{1 - \frac{2GM}{rc^2}}, \qquad \left(\frac{d\tau}{dt}\right)_{\text{circular}} = \sqrt{1 - \frac{3GM}{rc^2}}$$

Here t is Schwarzschild coordinate time referenced to infinity; a held clock requires support. The circular-clock formula includes orbital motion. Rotation, oblateness, other bodies and signal propagation are omitted. The website's spatial embedding has labelled vertical magnification; its apparent downward direction is not a physical extra direction [7].

6. How to explore, reproduce and critique

A short simulation tutorial

1. Start with the scales. In Milky Way, compare 100 years with 250 million years. A century is too short to reveal visible travel at the full Galactic scale.

2. Resolve 100 light-years. Open the journey preview. It takes about 128,608 model years to cover 100 ly. Record 10 seconds captures the live local scene; keep the tab visible.

3. Change the frame. Open Sun & all planets, use the winding-path tour, and compare full Galactic and 98% co-moving frames. Use the Moon close-up separately.

4. Inspect physical readouts. In Space & Time, compare Earth at 400 km and 20,200 km. Change diagram height and verify that numerical clock rates do not change.

5. Check the evidence. Use the report's source links, CSV exports and frozen inputs. The repository supplies numerical modules and validation scripts.

Questions for the community

A feasible next test. Which known planetary systems have orbital planes and relative velocities sufficiently constrained to test entry into a transit band?

A realistic observation model. Which apparent accessibility gains survive exposure-time budgets, repeated visits, binaries and instrument noise?

A second observer. What physically feasible probe orbit would produce an additional useful baseline beyond Earth's orbit and long-term astrometry?

Relevant prior work. Which studies or follow-up results best connect changing sightlines, nearby encounters and exoplanet detection opportunities?

I am an independent researcher without a formal supervisor. I built the software out of curiosity and would welcome expert guidance on what is useful, what is already established, and what needs correction.

Selected references and figure provenance

[1] ESA/Gaia/DPAC, Stefan Payne-Wardenaar (2025). Milky Way artist reconstruction.

[2] Kaltenegger & Faherty (2021). Past, present and future stars that can see Earth as a transiting exoplanet.

[3] Kervella, Thévenin & Lovis (2017). Proxima / Alpha Centauri input measurements.

[4] Reylé et al. (2021). The 10 parsec sample in the Gaia era.

[5] Juri■ et al. (2008). The Milky Way tomography with SDSS.

[6] Põder et al. (2023). Galactic circular-speed inference.

[7] Ashby (2003). Relativity in the Global Positioning System.

[8] Gaia DR3 data release; catalogue selection and uncertainty details in the full report.

[9] Stark et al. (2024). Robust HWO exoplanet yield margins.

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Simulation, full methods, original note and reproducible source:
biswajit1999.github.io/moving-observer/